

Airflow Analysis of an Aircraft Engine Axial Compressor with Smooth and Axial Groove Casing Using CFD

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ABSTRACT Transonic axial compressors operate close to their stability limits due to complex flow phenomena such as shock–boundary layer interaction and tip leakage vortices, which strongly influence performance and stall behavior. In the present work, a computational fluid dynamics analysis is performed on the NASA Rotor-37 to validate the numerical approach against available experimental data. The baseline rotor is simulated using the Reynolds-Averaged Navier–Stokes equations coupled with the SST $k - \omega$. Overall compressor performance characteristics, including pressure ratio, mass flow rate, efficiency, and spanwise distributions of total pressure and total temperature, are analyzed at near-peak efficiency and near-stall operating conditions. The numerical results show good agreement with experimental trends, while discrepancies near the hub and casing regions are observed, which are consistent with previously reported literature. After successful validation of the baseline model, axial grooves are introduced on the casing as a passive flow control technique to improve compressor stability. The grooved casing configuration effectively delays flow separation and weakens tip leakage vortex interaction, resulting in a reduction in stall tendency. As a result, an approximately 9.2% improvement in stall margin is achieved, leading to an extended stable operating range compared to the smooth casing. The study demonstrates that axial groove casing treatment is an effective method for enhancing the stability of transonic axial compressors.

Index Terms— *Transonic axial compressor, NASA Rotor-37, Computational fluid dynamics (CFD), SST $k - \omega$ turbulence model, Compressor stall, Axial groove casing treatment, Stall margin improvement*

I. INTRODUCTION

Axial compressors are critical components of modern gas turbine engines and aircraft propulsion systems, where high pressure ratio and compact design are required. Transonic axial compressors, in particular, operate under severe aerodynamic conditions due to the presence of shock waves, strong adverse pressure gradients, and tip leakage vortices. These complex flow phenomena significantly influence compressor efficiency, pressure rise, and stability, often leading to stall and flow separation near the operating limits. Accurate prediction of such behavior remains a major challenge in compressor design.

NASA Rotor-37 has been widely adopted as a benchmark test case for validating computational fluid dynamics (CFD) methods applied to transonic compressor flows. Extensive experimental data are available for this rotor, including overall performance characteristics and spanwise distributions of aerodynamic parameters, making it suitable for assessing the reliability of numerical models. However, previous studies have reported persistent discrepancies between CFD predictions and experimental results, particularly near the hub and casing regions, highlighting

the limitations of turbulence modeling and numerical approaches.

In this work, a CFD-based investigation of NASA Rotor-37 is carried out to validate the numerical methodology against published experimental data. Following successful validation, a passive flow control strategy in the form of axial groove casing treatment is introduced to improve compressor stability. The primary objective of this study is to evaluate the effectiveness of axial grooves in reducing stall tendency and enhancing the stable operating range of a transonic axial compressor.

Despite significant advancements in numerical methods and computational resources, accurate prediction of stall inception and stability enhancement techniques remains an active area of research. Passive flow control methods, such as casing treatments, have gained attention due to their simplicity and effectiveness without requiring external energy input. Among these, axial grooves on the casing have shown potential in modifying tip leakage flow behavior and delaying flow separation. However, the

effectiveness of such techniques strongly depends on reliable baseline validation. Therefore, establishing confidence in the CFD model through comparison with well-documented experimental data is essential before introducing design modifications. This study emphasizes a structured approach by first validating the baseline NASA Rotor-37 configuration and subsequently assessing the impact of axial groove casing treatment on compressor stall characteristics and overall performance.

II. LITERATURE REVIEW

Extensive research has been carried out on the numerical investigation and performance enhancement of transonic axial compressors due to their critical role in aerospace and gas turbine applications. Among the various benchmark configurations, NASA Rotor-37 has been widely studied because of the availability of comprehensive experimental data and its relevance to high-speed compressor stages. Previous studies have focused on understanding shock–boundary layer interactions, tip leakage vortices, and their influence on compressor efficiency and stall behavior.

A. NASA Rotor-37 CFD Validation Studies

Several researchers have utilized NASA Rotor-37 as a reference case to validate computational fluid dynamics (CFD) methodologies. Boretti [2] performed experimental and numerical investigations using RANS-based CFD with the SST $k - \omega$ turbulence model, comparing performance maps and spanwise distributions of total pressure and temperature with experimental data. The study reported good agreement in overall performance trends while highlighting discrepancies near the hub and casing regions, attributed to complex flow physics and turbulence modeling limitations.

AGARD and ASME blind test cases further established Rotor-37 as a standard validation problem for turbomachinery CFD. These studies demonstrated that although most CFD codes could capture global compressor characteristics, accurate prediction of hub pressure deficit and casing temperature rise remained challenging. Such findings emphasized the importance of careful model validation before applying CFD for design modifications.

B. Casing Treatment and Stall Control Techniques

To improve compressor stability, various passive flow control methods have been investigated, including circumferential grooves, axial grooves, and slot-type casing treatments. Ahmad et al. [3] numerically studied the effect of casing grooves on Rotor-37 and reported a significant improvement in stall margin with a minor reduction in efficiency. Their results indicated that casing grooves effectively weaken tip leakage vortex strength and delay flow separation near stall conditions.

Other studies have shown that groove geometry, axial location, and depth play a crucial role in determining the effectiveness of casing treatments. While circumferential grooves are widely studied, axial grooves have gained attention due to their ability to redistribute momentum near the blade tip region and suppress flow instabilities.

C. Research Gap and Present Work Motivation

From the literature, it is evident that most studies focus either on CFD validation or on casing treatment-based performance enhancement. However, limited work systematically combines baseline CFD validation of NASA Rotor-37 followed by the introduction of axial groove casing treatment using the same validated model. Therefore, the present work aims to first validate the CFD methodology against experimental data and subsequently assess the effectiveness of axial grooves in reducing stall tendency and improving compressor stability.

II. MATERIALS AND METHODS

The methodology adopted in this study is based on a computational fluid dynamics (CFD) approach combined with validation against published experimental data. Rather than experimental testing, the work focuses on numerical simulation, literature-based benchmarking, and analytical comparison to evaluate compressor performance and stability characteristics. The study follows a structured process involving baseline model validation of NASA Rotor-37 and subsequent introduction of axial groove casing treatment to assess its influence on stall behavior. This approach enables systematic analysis of flow physics, performance trends, and stability enhancement without physical prototyping.

A. Compressor Geometry and Reference Model (Literature-Based Configuration)

The compressor geometry used in this study is based on the well-documented NASA Rotor-37, which serves as a standard benchmark for transonic axial compressor analysis. Geometric parameters, including blade profile, hub-to-tip ratio, tip clearance, and operating speed, are adopted directly from published experimental literature. The rotor consists of 36 blades designed for transonic operation, and extensive experimental performance data are available for validation. Using literature-defined geometry ensures consistency with previous studies and enables reliable comparison of numerical results.

Fig. 1. CAD Model of NASA rotor 37

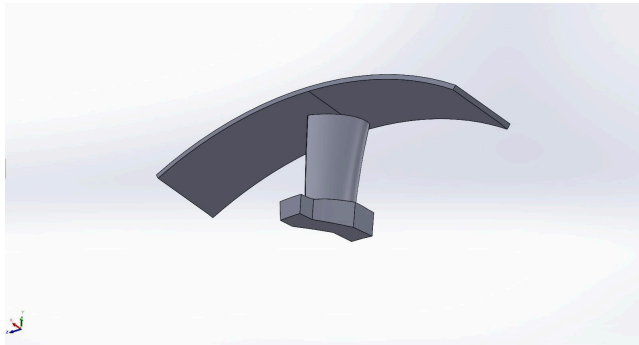


TABLE 1: Design Specifications Obtained from A.Boretti[2]

PARAMETER	DESIGN VALUE
Rotor Total Pressure Ratio	2.106
Rotor Total Temperature Ratio	1.270
Rotor Adiabatic Efficiency	0.877
Rotor Head Rise Coefficient	0.333
Flow Coefficient	0.453
Mass Flow, kg/s	20.188
Rotor Wheel Speed, rpm	17188.7
Rotor Tip Speed, m/s	454.14
Hub / Tip Radius Ratio	0.70
Rotor Aspect Ratio	1.19
Number of Rotor Blades	36
Blading Type	Multiple Circular Arc (MCA)

B. Numerical Modeling and Solver Setup

Boundary Name	Type	Settings
Inlet	Inlet	Total Pressure: 101,325 Pa (1 atm) Total Temperature: 288.15 K Flow Direction: Normal to Boundary
Outlet	Outlet	Average Static Pressure: Variable (We gradually increased this from 100 kPa to stall point to find the limit).
Blade & Hub	Wall	No Slip, Adiabatic, Rotating with Domain
Shroud (Smooth)	Wall	No Slip, Adiabatic, Counter-Rotating (Stationary in absolute frame)
Periodic Sides	Interface	Rotational Periodicity (Connecting the sides of the 10° wedge)

TABLE 2: Boundary Conditions

The flow field is simulated using the Reynolds-Averaged Navier–Stokes (RANS) equations to model compressible, turbulent flow through the compressor passage. The SST $k - \omega$ turbulence model is employed due to its proven capability in capturing adverse pressure gradients and separating flows common in transonic compressors. A single blade passage computational domain with periodic boundary conditions is considered to reduce computational cost while preserving flow physics. Inlet total pressure and total temperature, along with outlet static pressure, are specified based on experimental operating conditions reported in the literature.

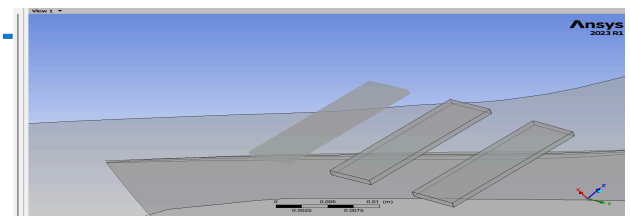
C. Baseline Validation Methodology

Baseline validation is performed by comparing numerical predictions with published experimental data for NASA Rotor-37. Key performance parameters such as pressure ratio, mass flow rate, efficiency, and spanwise distributions of total pressure and total temperature are evaluated at near-peak efficiency and near-stall operating conditions. The validation focuses on reproducing overall performance trends as well as known discrepancies near the hub and casing regions. This step establishes confidence in the numerical setup before introducing design modifications.

D. Axial Groove Casing Treatment and Analysis Workflow

Following baseline validation, axial grooves are introduced on the casing as a passive flow control modification. The grooved configuration is analyzed under identical boundary conditions to isolate the effect of axial grooves on compressor performance and stability. Comparative analysis is carried out between the smooth casing and grooved casing configurations, with emphasis on stall margin, pressure ratio variation, and flow stability. The overall workflow consists of geometry definition, mesh generation, baseline simulation, validation, axial groove implementation, and performance comparison. This structured methodology allows clear identification of the impact of axial groove casing treatment on stall reduction and stable operating range extension.

Fig. 2. Image of Axial Slots on the casing



III. RESULTS AND DISCUSSIONS

A. Comparative Analysis of Baseline and Modified Configurations

The CFD results obtained for the baseline NASA Rotor-37 configuration demonstrate trends consistent with previously reported experimental and numerical studies. The baseline simulations successfully capture key compressor performance characteristics, including choke, peak efficiency, and near-stall operating conditions. As expected from literature, the smooth casing configuration exhibits increasing pressure ratio with decreasing mass flow, followed by a rapid onset of stall at low flow rates. Minor discrepancies are observed near the hub and casing regions, particularly in spanwise total pressure and temperature distributions, which are consistent with known limitations of RANS-based turbulence models in predicting complex shock–boundary layer and tip leakage vortex interactions.

Upon introducing axial groove casing treatment, a noticeable improvement in compressor stability is observed. Compared to the smooth casing, the grooved configuration delays the onset of stall and allows stable operation at lower mass flow rates. This behavior indicates effective modification of near-tip flow structures and redistribution of momentum, confirming the beneficial role of axial grooves in stabilizing transonic compressor flow.

TABLE 3: Performance characteristics of smooth casing

Back Pressure (Pa)	Mass in (Kg/s)	Mass out (Kg/s)	P ratio	T ratio	Efficiency	inference
101017	20.981	20.981	1.814	1.221	0.839	choked
101018	20.981	20.981	1.814	1.221	0.839	
115775	20.93	20.924	1.964	1.2428	0.859	
125900	20.805	20.791	2.076	1.269	0.862	Near peak
130916	20.406	20.402	2.123	1.28	0.857	
130922	20.435	20.428	2.124	1.28	0.858	
135850	19.432	19.429	2.158	1.294	0.836	near st
139162	8.316	7.795	1.845	1.251	0.761	

TABLE 4: Performance characteristics of Axial groove on casing

#	Back Pressure (Pa)	#	Mass in (Kg/s)	#	Mass out (Kg/s)	#	P ratio	#	T ratio	#	Efficiency	inference
	100931		20.889		20.886		1.818		1.225		0.8227	choked
	110727		20.788		10.782		1.914		1.244		0.836	
	115716		20.726		20.724		1.965		1.253		0.841	peak
	121219		20.429		20.43		2.013		1.264		0.837	
	125611		19.55		19.544		2.027		1.273		0.819	
	126111		19.505		19.504		2.23		1.347		0.742	near stall
	134396		8.454		8.049		1.833		1.257		0.735	

B. Performance and Stability Characteristics

B.1. Stall Margin Improvement

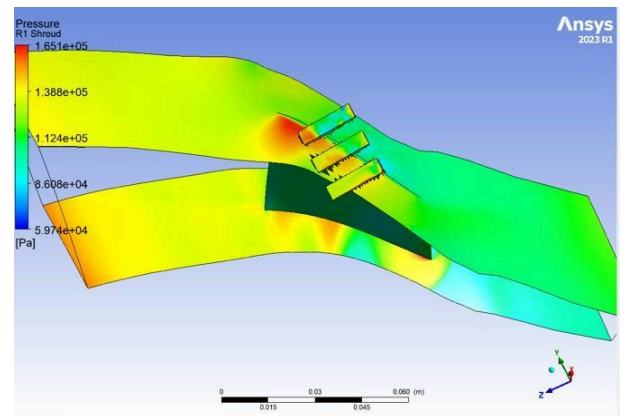
Comparative analysis of performance curves shows that the axial groove configuration achieves an approximately **9.2% improvement in stall margin** relative to the baseline smooth casing. The stall point is shifted toward lower mass flow rates, indicating enhanced resistance to flow separation and instability. This improvement is attributed to the interaction between the axial grooves and tip leakage flow, which weakens vortex strength and delays vortex breakdown near stall conditions.

C. Flow Physics and Practical Implications

The observed improvement in stall margin can be linked to altered flow behavior near the casing region. Axial grooves promote momentum exchange between the main flow and the tip leakage region, thereby reducing flow blockage and suppressing early separation. This stabilization mechanism is particularly effective under near-stall operating conditions, where shock–vortex interactions dominate flow behavior.

From an application perspective, the results demonstrate that axial groove casing treatment is a viable passive flow control technique for enhancing the stable operating range of transonic axial compressors. The CFD-based findings support the potential use of such modifications in compressor design, provided that baseline validation is performed to ensure reliability of numerical predictions.

Fig. 3. The Pressure gradient in the axial slots



IV. CHALLENGES AND LIMITATIONS

A. Technical Challenges

1. Turbulence Modeling Limitations

The accuracy of CFD predictions for transonic axial compressors is strongly influenced by the choice of turbulence model. Although the SST $k - \omega$ model provides reliable prediction of adverse pressure gradients and near-wall flows, it has known limitations in accurately capturing complex shock–boundary layer interactions and tip leakage vortex behavior. As a result, discrepancies persist in hub pressure deficit and casing temperature prediction, as also reported in existing literature.

2. Sensitivity to Numerical and Boundary Conditions CFD results for NASA Rotor-37 are highly sensitive to mesh resolution, near-wall treatment, and outlet boundary conditions. Small variations in mesh density or pressure outlet specification can lead to noticeable changes in predicted mass flow, pressure ratio, and stall point. This sensitivity increases computational effort and requires careful numerical setup to ensure consistent and repeatable results.

3. Steady-State Approximation

The present study employs a steady-state RANS approach, which may not fully capture unsteady flow phenomena such as rotating stall inception and transient vortex breakdown. While steady simulations are effective for predicting overall performance trends, they limit the ability to resolve time-dependent instability mechanisms in detail.

B. Scalability and Applicability Considerations

1. Computational Cost and Resource Requirements High-fidelity CFD simulations of transonic compressors require fine meshes and extended iteration counts to achieve convergence, resulting in significant computational cost. This limits the ability to perform extensive parametric studies involving multiple groove geometries and operating conditions.

2. Simplified Geometry Representation:

The axial groove casing treatment implemented in this study is based on simplified geometric assumptions. In practical compressor designs, manufacturing constraints and structural considerations may influence groove shape, depth, and placement, potentially affecting performance outcomes.

3. Lack of Experimental Validation for Modified Configuration

While the baseline NASA Rotor-37 model is validated against experimental data, the grooved casing configuration lacks direct experimental validation. Therefore, the observed stall margin improvement is based solely on

numerical evidence, and experimental testing would be required to confirm the effectiveness of axial grooves under real operating conditions.

V. CONCLUSIONS

In this study, a computational fluid dynamics (CFD) investigation of the NASA Rotor-37 transonic axial compressor was carried out to validate the numerical methodology and to assess the effectiveness of axial groove casing treatment for stall reduction. The baseline smooth casing configuration was first simulated using the Reynolds-Averaged Navier–Stokes equations with the SST $k - \omega$ turbulence model. The numerical results showed good agreement with published experimental data in terms of overall performance characteristics, including pressure ratio, mass flow rate, efficiency, and spanwise distributions of total pressure and total temperature. The observed discrepancies near the hub and casing regions were consistent with those reported in existing literature, confirming the reliability of the validated CFD model.

Following validation, axial grooves were introduced on the casing as a passive flow control technique. The modified configuration demonstrated a clear improvement in compressor stability, with an approximately 9.2% increase in stall margin compared to the smooth casing. The pressure ratio and efficiency trends near the design operating point remained largely unaffected, indicating that the stability enhancement was achieved with minimal performance penalty. The results confirm that axial groove casing treatment is an effective and practical approach for extending the stable operating range of transonic axial compressors when applied using a validated numerical framework.

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VII. REFERENCES

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